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$^{+4}$Notes of Complex Analysis

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"La calle es pa' hombres, por eso está la acera" Nestor Heli Aponte Avila[1]
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** Content based on the course MM413 (Variavél Complexa) taught by Professor Tiago Jardim Fonseca in the semester 2026-I, mainly following [1]. **

Notation ◇ Definition $(-)^{\diamond}$ Open $(-)^{\blacklozenge}$ Closed
□ Lemma ☐ Proposition ■ Theorem ⊠ Corollary

§ 1 Preliminaries

Let $\mathbb{C} := \mathbb{R}[i]$ the extension field of $\mathbb{R}$, obtained by adjoining i, the *imaginary unity*, which satisfies $i^2 = -1$. Some important details about $\mathbb{C}$:

- Any $z \in \mathbb{C}$ can be written as $z = x + iy$, where $x, y \in \mathbb{R}$ are the *real* and *imaginary parts* of z. Also, $\overline{z} := x - iy$ denotes the *complex conjugate* of z.
$$x = \operatorname{Re}(z) = \frac{z + \overline{z}}{2} \quad \text{and} \quad y = \operatorname{Im}(z) = \frac{z - \overline{z}}{2i}.$$
- $\mathbb{C} \simeq \mathbb{R}^2$, so there are natural geometric interpretations for the operations on complex numbers.
- The map $(z, w) \mapsto \langle z, w \rangle := z\overline{w}$ is a Hermitian inner product and induces the norm $|z|^2 = z\overline{z} = x^2 + y^2$ (the same as the usual for $\mathbb{R}^2$). Useful inequalities:
  i. $|z + w| \leq |z| + |w|$. ii. $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$ (em $\mathbb{C}^\times$).
  iii. $||z| - |w|| \leq |z - w|$. iv. $|\operatorname{Re}(z)|, |\operatorname{Im}(z)| \leq |z|$.
- We can write $z \in \mathbb{C}^\times$ in *polar form* as $z = r\exp(i\theta) = r\operatorname{cis}(\theta)$, where $\theta = \arg(z)$ is the *argument* of z and $r = |z| > 0$ is the modulus.

Following the standard topology notation, given $z_0 \in \mathbb{C}$ and $r > 0$, $B(z_0; r) := \{z \in \mathbb{C} : |z - z_0| < r\}$, $B[z_0; r] := \overline{B(z_0; r)}$ and $\partial B(z_0; r) = \{z \in \mathbb{C} : |z - z_0| = r\}$.

$$\diamondsuit \quad \mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}.$$

Naturally, $\mathbb{C}$ is a metric space and we work with the topology induced by the metric. In this context, open sets are completely characterized by the basis $\mathcal{B} := \{B(z_0; r) : z_0 \in \mathbb{C},\ r > 0\}$, along with all the associated facts and theory (compactness and connectedness in particular).

$$\diamondsuit \quad \operatorname{diam} \Omega := \sup_{\Omega} |z - w|.$$

Holomorphic functions

$\diamondsuit$ A function $f : \Omega^{\diamond} \subset \mathbb{C} \to \mathbb{C}$ is *holomorphic* at $z_0 \in \Omega$ if the limit,

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}, \qquad !!\ h = h_1 + ih_2 \in \mathbb{C}$$

exists. When it exists, $f'(z_0)$ is called the *derivative* of f at z_0. We say holomorphic in Ω if $\forall z_0 \in \Omega,\ \exists f'(z_0)$, and *entire* if it is holomorphic on all $\mathbb{C}$.

Example Any polynomial function $p \in \mathbb{C}[z]$ is entire and $1/z$ is holomorphic on $\mathbb{C}^{\times}$. On the other hand, the conjugation $z \mapsto \overline{z}$ is not holomorphic. $\to$ Make the limit

Note. An equivalent and useless formulation for holomorphic at z_0 is that $\exists \alpha \in \mathbb{C} : f(z_0 + h) = f(z_0) + \alpha \cdot h + o(h)$, of course that $\alpha = f'(z_0)$.

$\square$ **2.2.** If f and g are holomorphic (in appropriate domains), then $f + g,\ fg,\ f/g$ and $g \circ f$ are holomorphic too. $\to$ Same formulas and proof of analysis in $\mathbb{R}^n$

$$F : \mathbb{R}^2 \to \mathbb{R}^2 \text{ diff } \not\Rightarrow F \sim f : \mathbb{C} \to \mathbb{C} \text{ homorphic}$$

Example $F : (x, y) \mapsto (x, -y) \sim f : z \mapsto \overline{z}$.

$$\diamondsuit \quad \partial_z := 1/2\,(\partial_x + 1/i\,\partial_y) \ \text{ and } \ \partial_{\overline{z}} := 1/2\,(\partial_x - 1/i\,\partial_y).$$

$\square$ **2.3.** If $f(z) = u(x, y) + iv(x, y)$ is holomorphic at $z_0 = x_0 + iy_0$, then

$$f_{\overline{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = f_z(z_0) = 2u_z(z_0).$$

Also, $F(x, y) = (u(x, y), v(x, y))$ is diff (real sense) and $\det JF(x_0, y_0) = |f'(z_0)|^2$.

For first equal (Cauchy-Riemann equations) take the limits,

$$\lim_{h_1 \to 0} \frac{f(z_0 + h_1)}{h_1} = f'(z_0) = \lim_{ih_2 \to 0} \frac{f(z_0 + ih_2)}{ih_2} \Rightarrow \begin{cases} u_x = v_y \\ u_y = -v_x \end{cases}.$$

Other affirmations are consequence of C-R equations,

$$JF(x_0, y_0)(h) = \begin{bmatrix} u_x & u_y \\ -u_y & u_x \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = (u_x - iu_y)(h_1 + ih_2) = f'(z_0) \cdot h,$$

also $\det JF(x_0, y_0) = u_x^2 + u_y^2 = |f'(z_0)|^2$.

■ **2.4.** Let $\Omega \simeq U^{\diamond} \subset \mathbb{R}^2$, $u, v \in C^1(U, \mathbb{R}^2)$ and $f = u + iv : \Omega \to \mathbb{C}$. If u and v satisfies the C-R equations, then f is holomorphic on Ω and $f'(z) = f_z$. $\to$ Write du and dv by extension and group them, see [1, pag. 13]

Power series

The most important examples are the exponential series, which converges absolutely on compact sets, and the geometric series, which converges absolutely on $\mathbb{D}$.

- $\exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$. $\to \left| \sum \frac{z^n}{n!} \right| \leq \sum \frac{|z|^n}{n!} = \exp|z| < \infty$
- $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$, $|z| < 1$. $\to S_N = \frac{1-z}{1-z} \sum^{N} z^n = \frac{1 - z^{N+1}}{1-z} \to \frac{1}{1-z}$

In general, a *power series* is a sum of terms of the form $(a_n z^n)_{n \geq 0}$ and we interest in its absolute convergence.

■ **2.5.** Given a power series $\sum a_n z^n$, there exists $R \in [0, \infty]$ such that:

$$\sum_{n=0}^{\infty} |a_n||z|^n = \begin{cases} L, & |z| < R \\ \infty, & |z| > R \end{cases}.$$

Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then R is given by *Hadamard's formula*: $1/R = \limsup |a_n|^{1/n}$.

Let $L := 1/R$, for $|z| < R$ chooses $0 < \epsilon \ll 1 : (L + \epsilon)|z| < r < 1$. Follows that $|a_n||z|^n \leq \{(L + \epsilon)|z|\}^n = r^n$, geometric series. Similiar argument concludes divergence for case $|z| > R$.

The case $|z| = R$ is not conclusive

Example sdfjksjs

Euler's formulas for sine and cosine

- $\cos(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = \frac{e^{iz} + e^{-iz}}{2}$
- $\sin(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = \frac{e^{iz} - e^{-iz}}{2i}$

■ **2.6.** The power series $f(z) = \sum a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of f is also a power series,

$$f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1} =: g(z) \text{ (termwise differentiation)},$$

with same radius of converges as f.

Remember that $n^{1/n} \to 1$ as $n \to \infty$, so from Hadamard's formula follows the affimation of radius of convergence of g. The first assertion doesn't admite summaries, see [1, Pág. 17]. Consists of an $\epsilon/3$ argument, using,

$$f(z) = S_N(z) + E_N(z) = \sum_{n=0}^{N} a_n z^n + \sum_{N+1}^{\infty} a_n z^n.$$

⊠ **2.7.** A power series is an infinitely differentiable in its disc of convergence, and the higher derivatives are also a power series obtained by termwise differentiation.

Note. The same holds for series centered at $z_0 \in \mathbb{C}^\times$, it's simply translations.

Integration along curves

◇ A *smooth parametrized curve* is a function $z : [a, b] \to \mathbb{C}$ such that $\exists z'(t) \neq 0$ continuous on $[a, b]$, with lateral limits for $t = a, b$.

◇ Two parametrizations $z : [a, b] \to \mathbb{C}$ and $\widetilde{z} : [c, d] \to \mathbb{C}$ are *equivalent*, denote $z \sim \widetilde{z}$, if $\exists\, {}^1t : [a, b] \simeq [c, d]$ such that $t'(s) > 0$ and $\widetilde{z}(s) = z(t(s))$. The clases $[z]$ determines a *smooth curve* $\operatorname{Im} z =: \gamma \subset \mathbb{C}$ with orientation $z(a) \rightsquigarrow z(b)$.

Note. It's possible to extend this definitions naturally for *piecewise-smooth curve*. We have particular interest in *closed curves* $z \mid z(a) = z(b)$.

Example $\partial B(z_0; r) = z_0 + re^{it},\ t \in [0, 2\pi]$ (positive orientation).

$\diamondsuit$ Given a smooth curve $\gamma \subset C$ parametrized by $z : [a, b] \to \mathbb{C}$, and f a continuous function on γ, we define,

$$\int_\gamma f(z)\, dz := \int_a^b f(z(t))z'(t)\, dt.$$

Note. It's easy to see that the integral is independent of the parametrization. $\to$ Just use an equivalent parametrization and chain's rule

$$\diamondsuit \ \text{length}(\gamma) := \int_a^b |z'(t)|\, dt.$$

$\square$ **3.1.** Integration of continuous functions on curves satisfies:

i. $\displaystyle\int_\gamma f(z) + \alpha g(z)\, dz = \int_\gamma f(z)\, dz + \alpha \int_\gamma g(z)\, dz.$

ii. $\displaystyle\int_\gamma f(z)\, dz = -\int_{\gamma^-} f(z)\, dz.$ iii. $\displaystyle\left|\int_\gamma f(z)\, dz\right| \leq \sup_{z\in\gamma} |f(z)| \cdot \text{length}(\gamma).$

■ **3.2.** Let $F : \Omega^\diamond \subset \mathbb{C} \to \mathbb{C}$ holomorphic, $\Omega \supset \gamma : w_1 \rightsquigarrow w_2$ smooth curve and $dF(z) = F'(z) = f(z)\, dz$. Then,

$$\int_\gamma f(z)\, dz = \int_{\partial\gamma} F(z) = F(w_2) - F(w_1).$$

It follows by ■ (*Stokes*) or ■ (*TFC*) if you prefer.

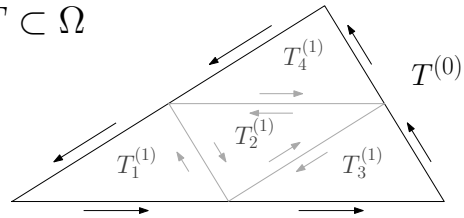
⊠ **3.3.** In the setting of ■ 3.3. if γ is a closed curve, then, $\displaystyle\int_\gamma f(z)\, dz = 0.$

⊠ **3.4.** If f is holomorphic on $\Omega^\diamond \subset \mathbb{C}$ connected and $f' \equiv 0$, then f is constant. $\to$ It suffices to take a smooth curve $\gamma : w_0 \rightsquigarrow w$ and apply ■ 3.3.

§ 2 Cauchy's Theorem and Its Applications

■ **1.1.** (***Goursat***) Let f holomorphic in $\Omega^{\diamond} \subset \mathbb{C}$ and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\, dz = 0.$$



→ Doesn't support summaries, but rather on a bisection argument, see [1, pag. 34]

⊠ **1.2.** If R is a rectangle : $R, \operatorname{int} R \subset \Omega$, then $\displaystyle\int_R f(z)\, dz = 0$.

■ **2.1.** If f is holomorphic in $D := B(z_0; r)$, then f has a primitive in D. → Doesn't support summaries, see [1, pag. 37]

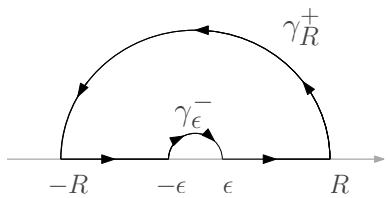
■ **2.2.** (***Cauchy's theorem for a disc***) If f is holomorphic in D, then

$$\int_\gamma f(z)\, dz = 0 \ : \ \gamma \subset D, \text{ closed curve.}$$

By ■ 2.1. f has a primitive, so from ⊠ 3.3. the result follows.

⊠ **2.3.** If f is holomorphic in Ω and $\overline{D} \subset \Omega$, then $\displaystyle\int_{\partial D} f(z)\, dz$.

Motivation behind theorem



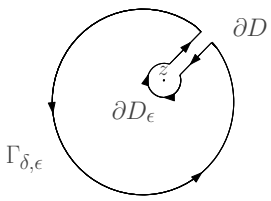
Example Evaluate integral using ■ 2.2., making $\epsilon \to 0$,

$$\frac{\pi}{2} = \int_0^\infty \frac{1 - \cos x}{x^2}\, dx \sim f(z) = \frac{1 - e^{iz}}{z^2}.$$

Cauchy's integral formula

■ **4.1.** If f is holomorphic in $\Omega^{\diamond} \subset \mathbb{C}$ and $D : \overline{D} \subset \Omega$, then

$$f(z) = \frac{1}{2\pi i} \int_{\partial D} \underbrace{\frac{f(\zeta)}{\zeta - z}}_{F(\zeta)}\, d\zeta, \quad z \in D.$$



$$0 = \lim_{\epsilon \to 0} \int_{\Gamma_{\delta,\epsilon}} F(\zeta)\, d\zeta = \int_{\partial D} F(\zeta)\, d\zeta + I_\epsilon + \int_{\partial D_\epsilon} \frac{f(z)}{\zeta - z}\, d\zeta.$$

$$I_\epsilon = \int_{\partial D_\epsilon} \frac{f(\zeta) - f(z)}{\zeta - z}\, d\zeta \to 0 \text{ and } \int_{\partial D_\epsilon} \frac{f(z)}{\zeta - z}\, d\zeta = -f(z) 2\pi i.$$

⊠ **4.2.** (***Cauchy integral formula***) If f is holomorphic in $\Omega^\diamond \subset \mathbb{C}$, then f has infinitely many complex derivatives in Ω. Moreover, if $\overline{D} \subset \Omega$, then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\partial D} \frac{f(\zeta)}{\zeta - z}\, d\zeta, \quad z \in D.$$

By induction, calculate $f^{(n)}(z) = \lim_{h \to 0}(\cdots)$ using ■ 4.1. to express $f^{n-1}(z)$ and the property $A^n - B^n = (A - B)[A^{n-1} + A^{n-2}B + \cdots + B^{n-1}]$ to simplify.

⊠ **4.3.** (***Cauchy inequalities***) If f is holomorphic in $\Omega^\diamond \subset \mathbb{C}$ and $B[z_0; R] \subset \Omega$, then

$$|f^{(n)}(z_0)| \leq \frac{n! \|f\|_{\partial D}}{R^n}, \text{ where } \|f\|_{\partial D} := \sup_{\partial D} |f(z)|$$

$$|f^{(n)}(z_0)| = \frac{n!}{2\pi} \left| \int_{\partial D} \frac{f(\zeta)}{\zeta - z}\, d\zeta \right| = \frac{n!}{2\pi} \left| \int_0^{2\pi} (\cdots)\, d\theta \right| \leq \frac{n!}{2\pi} \frac{\|f\|_{\partial D}}{R^n} 2\pi.$$

■ **4.4.** If f is holomorphic in $\Omega^\diamond \subset \mathbb{C}$ and $D = B[z_0; R] \subset \Omega$, then f has a convergent Taylor power series expansion at z_0:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \text{ where } a_n = \frac{f^{(n)}(z_0)}{n!}, \ z \in D.$$

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \frac{1}{\zeta - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0}\right)^n.$$

The geometic series converges because $|z - z_0| / |\zeta - z_0| < r < 1$ (uniformity). Finally, use ■ 4.2. (*Cauchy integral formula*) to conclude.

⊠ **4.5.** (***Liouville***) If f is entire and bounded, then f is constant. $\to$ By ⊠ 4.3. (*Cauchy inequalities*) $|f'(z_0)| \leq M/R \to 0$ as $R \to \infty$, so $f'(z) \equiv 0$, result follows by ⊠ 3.4.

⊠ **4.7.** Every $P(z) \in \mathbb{C}[z] \setminus \mathbb{C}$ has precisely n roots in $\mathbb{C}$ (algebraically closed).

First prove that $P(z)$ has one root, if not $1/P(z)$ is entire and bounded, then constant, ⊠ 4.5. (*Liouville*).↯ Result follows by induction and division algorithm.

■ **4.8.** Suppose that f is holomorphic in $\Omega^{\diamond} \subset \mathbb{C}$ connected : f vanishes on a sequence of a distinct points with a limit point in $z_0 \in \Omega$. Then, $f \equiv 0$ in Ω.

Prove first by contradiction that $f(z) = \sum a_n(z - z_0)^n \equiv 0$ on $B(z_0; \epsilon)$. Therefore, $\{z \in \Omega : f(z) = 0\} =: U = \overline{U} = U^{\circ} \leftrightharpoons U = \Omega$.

⊠ **4.9.** If f and g are holomorphic in $\Omega^{\diamond} \subset \mathbb{C}$ and $\varnothing \neq \{z \in \Omega : f(z) = g(z)\}^{\diamond} \subset \Omega$, then $f \equiv g$ throughout Ω.

Note. If f and F are holomorphic in Ω and Ω' respectively, with $\Omega \subset \Omega'$ and $f \equiv F$ in Ω, then F is a (unique) *analytic extension* of f from Ω to Ω'.

References

[1] Stein, E. M., & Shakarchi, R. (2010). *Complex analysis (Vol. 2)*. Princeton University Press.